Q8. Search Space and a Multi-Agent System that Combines Two Drawing Patterns

> **Question.** What is a search space? Design a multi-agent system that combines two drawing patterns into a new creative drawing, by expanding the search space or adding a new action. How can we combine two actions? Both agents introduce new combinations to create new states. What could the fitness function be?

1. The idea in plain words

Think of two artists. One only draws square spirals; the other only draws stars. Each artist knows a few moves: go forward, turn, make the next step longer. Neither can draw anything new with their own moves alone. The trick is to **glue one move from each artist into a brand-new move**, for example "turn like the star *and* grow like the spiral" in a single step. That new move opens up drawings neither artist could reach before. Two agents keep proposing such combinations, a fitness function scores them, and a shared archive remembers what is already done.

2. What is a search space?

The **search space** is the set of all states an agent can reach from the start state by applying its actions (operators), again and again. It is an implicit graph: states are nodes and actions are edges. We never build it fully; the search algorithm explores a small part of it. Two things decide its size:

- **The action set.** More actions means more edges out of every state.
- **The depth.** With a actions and sequences of length L, there are a^L possible sequences.

Here a state is a **drawing program** (a motif of actions repeated n times). Adding new actions therefore expands the search space directly. That is the creative lever in this question.

3. Assumptions

- Drawings are made by a "turtle" pen. The basic actions are `F` (forward), `L(θ)` (turn left), `G(g)` (make the step longer) and `C` (change colour).
- **Agent A** owns the square spiral: repeat [F, L(90°), G(3)] 40 times.
- **Agent B** owns the five-point star: repeat [F, L(144°)] 5 times.
- The agents work in rounds (6 rounds, 5 proposals each per round) and share one archive of "already seen" drawings.

4. Design of the multi-agent system

Part	In this design
Agents	A-spiral (likes compose, blend, nest) and B-star (likes compose, interleave, nest)
State	A drawing program: one motif of actions repeated n times (n = 20–70)
Basic actions	F, L, G, C (4 actions)
New actions	`LG` = turn + grow in one move, `STAR` = whole star as one move, `BLEND` = mixed turn angle (now 7 actions)
Agent behaviour	60% of the time an agent tweaks its own best idea (exploit); otherwise it tries a fresh combination (explore)
Shared archive	Stores every drawing that is clearly new (overlap with all archived drawings below 60%)
Goal	Highest fitness after 6 rounds, plus a set of diverse new drawings in the archive

5. How can we combine two actions?

Way	What it means	Example from our run
Compose	Run one action right after the other and treat the pair as one new action: $a_2 \circ a_1$	LG = L(144°) from the star + G(3) from the spiral. The result is a growing star-spiral.
Blend	Mix the parameters of two similar actions	BLEND turn = $\alpha \cdot 90° + (1-\alpha) \cdot 144°$. With $\alpha = 0.5$ that is 117°.
Nest	Pack one whole pattern into a single action and use it inside the other	STAR is drawn at every corner of the spiral (two levels of scale).
Interleave	Alternate the two parents' moves	One spiral move, then one star move, and so on.

Every way creates a **new action**. The action space grows from 4 to 7. For a motif of 6 actions the search space grows from 4^6 = 4,096 to 7^6 = 117,649 programs: **28.7 times larger** (Figure 3, right). This is exactly how adding an action expands the search space.

6. Step-by-step approach

1. Give each agent its parent pattern and the four basic actions.
2. In each round, every agent picks one action from each parent and combines them (compose, blend, nest or interleave).
3. Repeat the new action n times to get a drawing, which is a new state.
4. Score it with the fitness function.
5. If it is clearly different from everything in the shared archive, add it. Both agents now avoid it.
6. Each agent keeps its best drawing and, next round, either tweaks it or tries a fresh combination.
7. After the last round, show the best drawings as *choices* to the human designer.

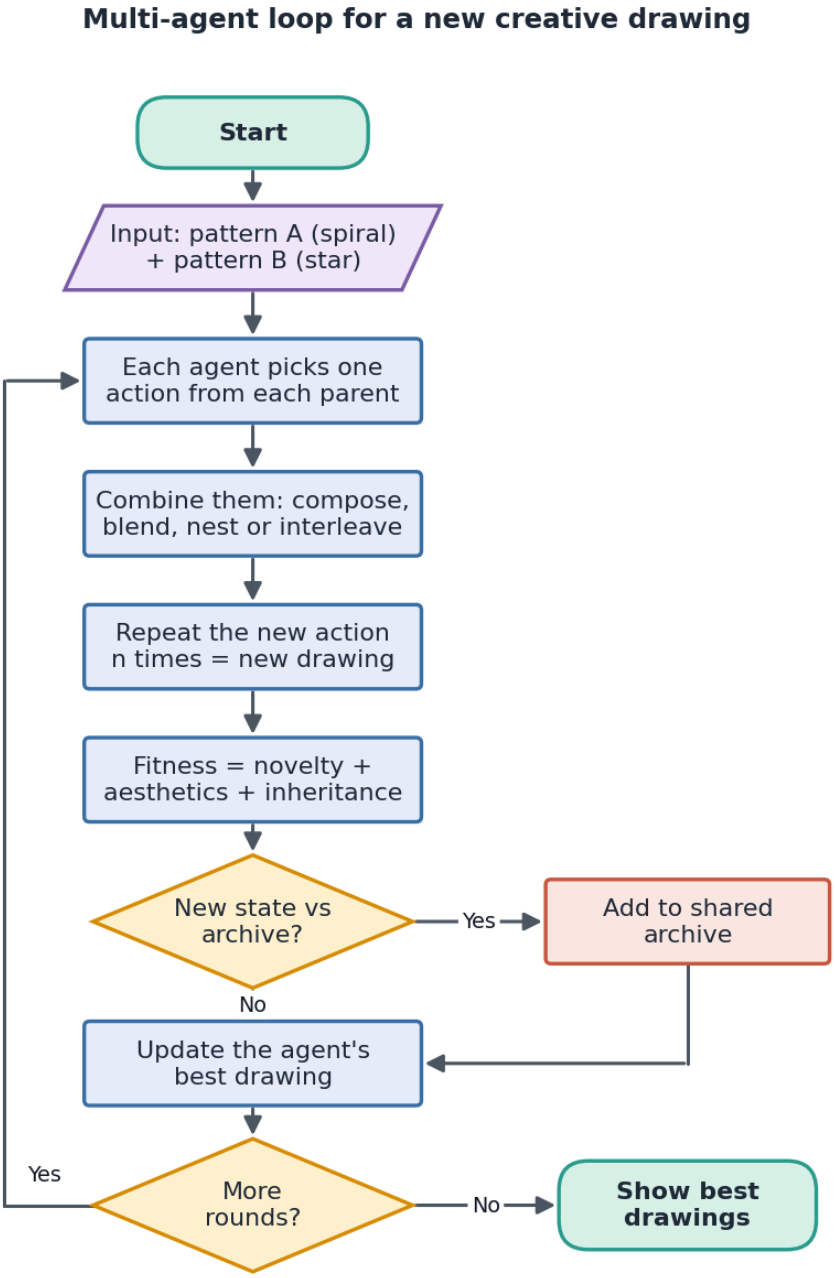


Figure 1. Flowchart of the multi-agent drawing system.

7. Pseudocode

```
function MultiAgentDrawing(parentA, parentB, rounds, k):
    archive ← { };  best[agent] ← none for both agents
    for r = 1 to rounds:
        for each agent in {A-spiral, B-star}:
            repeat k times:
                if best[agent] exists and random() < 0.6:
                    candidate ← combine(best[agent].operator, mutate(best[agent].params))
                else:
                    op ← random choice from agent's favourite operators
                    candidate ← combine(op, random params)          # new action
                drawing ← run_turtle(candidate)                         # new state
                score ← fitness(drawing)
                if overlap(drawing, every archived drawing) < 0.6: archive.add(drawing)
                if score > best[agent].score: best[agent] ← candidate
    return best drawings + archive
```

8. What could the fitness function be?

Fitness = 0.4 × Novelty + 0.4 × Aesthetics + 0.2 × Inheritance

- **Novelty** = 1 − the largest overlap (IoU of the pixel grids) with either parent. A drawing that looks like a plain spiral or a plain star scores low.
- **Aesthetics** borrows from Alexander's *living geometry*. It combines:
  - **local symmetry** (does the drawing match itself after a 3-, 4-, 5-, 6- or 8-fold rotation?);
  - **balanced coverage** (about 30% of the canvas inked);
  - **levels of scale** (a healthy mix of short and long strokes).
- **Inheritance** = $1 - |2 \cdot \text{share}_A - 1|$. It is 1.0 when both parents contribute equally, so the child is a real *combination* and not a copy of one parent.

9. Worked example (real run)

Both parents start with an aesthetics score of only **0.540** (spiral) and **0.547** (star). The table shows each agent's best drawing so far after every round:

Round	Agent	Best operator	Fitness	Novelty	Aesthetics	Archive size
1	A-spiral	compose	0.703	0.732	0.525	2
1	B-star	nest	0.616	0.499	0.641	3
3	A-spiral	compose	0.779	0.684	0.763	10
3	B-star	compose	0.735	0.677	0.661	13
6	**A-spiral**	**compose**	**0.793**	**0.672**	**0.810**	**23**
6	B-star	compose	0.749	0.669	0.703	26

Check of the winner: 0.4 × 0.672 + 0.4 × 0.810 + 0.2 × 1.0 = **0.793**. B-star started with "nest", then switched to "compose" after it saw that operator do better. By the end the archive held **26 different new drawings**.

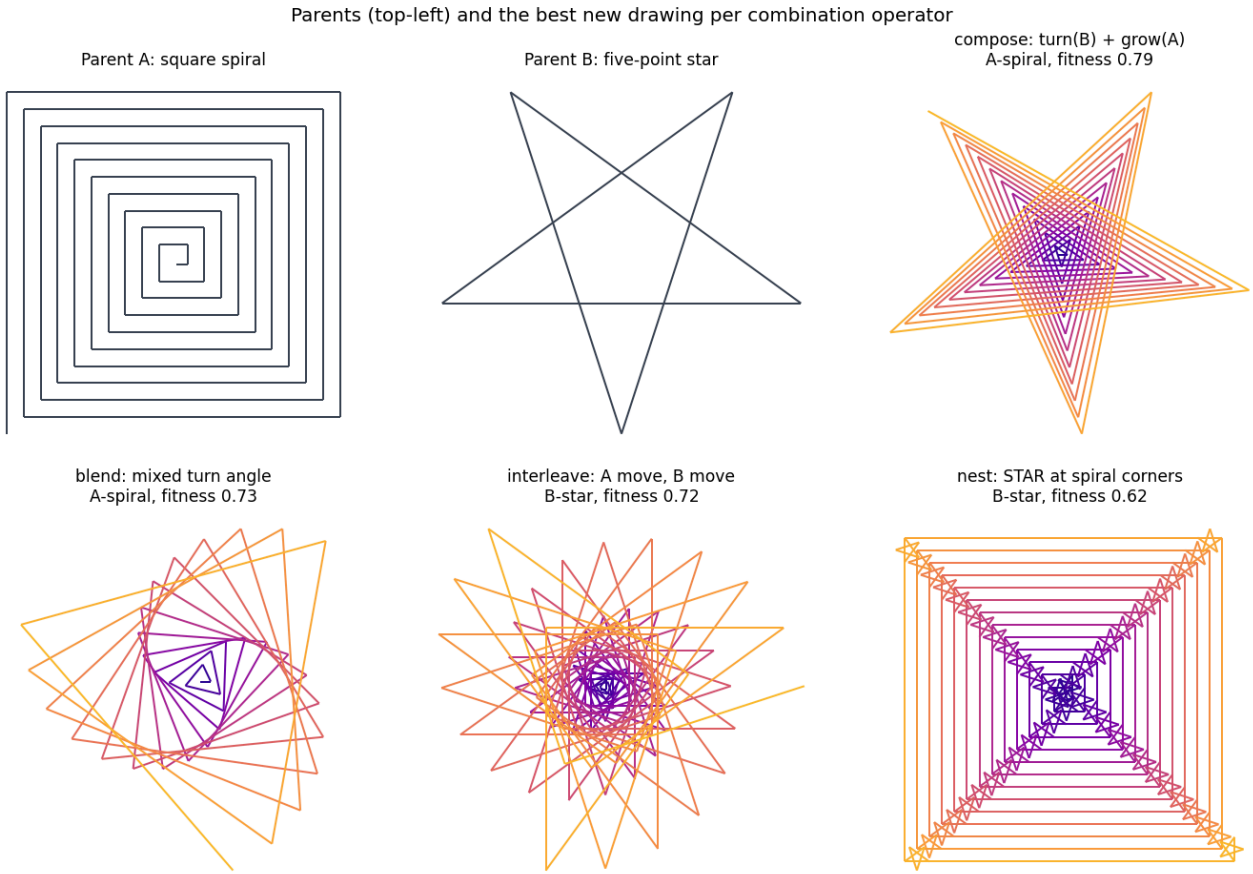


Figure 2. The real drawings. Top-left: the two parents. Others: the best new drawing for each way of combining actions.

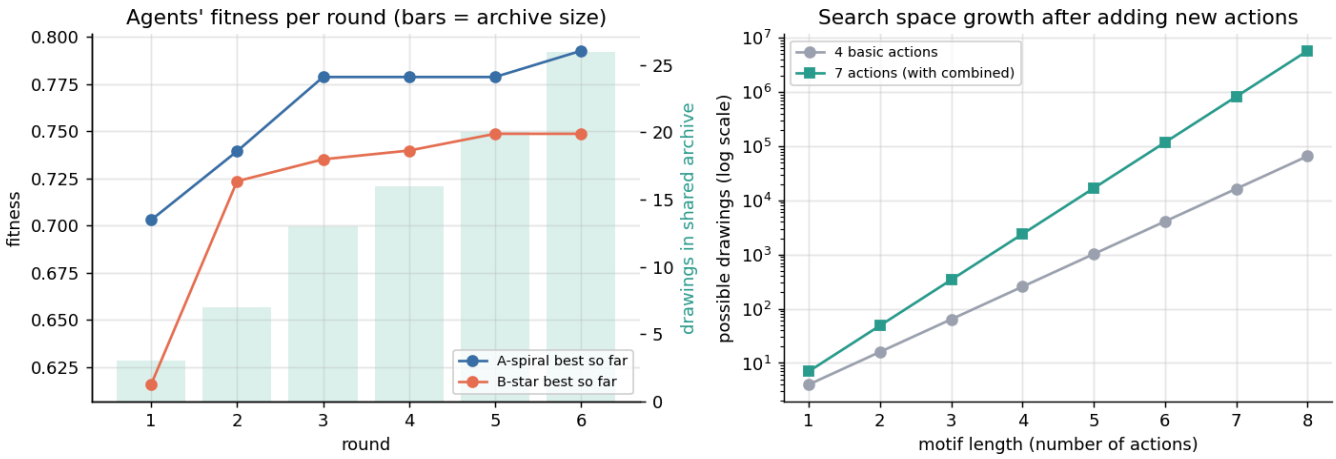


Figure 3. Left: each agent's best fitness per round, with archive size as bars. Right: search-space growth after adding new actions.

10. How do both agents create new states together?

- **Different styles.** A-spiral prefers blend; B-star prefers interleave. Their proposals come from different corners of the search space, which keeps diversity high. This is co-operative creativity, not two copies of one agent.
- **Shared archive.** Once one agent adds a drawing, the other gets no credit for repeating it. The pair is pushed toward fresh states.
- **Learning from each other.** Both agents ended up favouring "compose", because the archive and scores showed it worked.
- The final archive is a set of **choices, not one answer**. A human artist can pick the one they like.

11. Code and real run

The program `q08_code.py` draws the parents, runs both agents for 6 rounds and saves every figure in this answer.

File Edit View Run Terminal Help — q08_code.py - Q08_MultiAgent_Drawing_Patterns - Python IDE

q08_code.py

```python
AGENTS = {"A-spiral": ["compose", "blend", "nest"], "B-star": ["compose", "interleave", "nest"]}


def multi_agent_search(rounds: int = 6, proposals: int = 5) -> tuple:
    archive, history, all_cands = [], [], []  # archive = shared memory of new states
    best = {a: None for a in AGENTS}
    for r in range(1, rounds + 1):
        for agent, ops in AGENTS.items():
            for _ in range(proposals):
                if best[agent] and random.random() < 0.6:  # exploit: tweak own best idea
                    c = combine(best[agent]["op"], mutate(best[agent]["params"]))
                else:  # explore: brand-new combination of the two parents
                    c = combine(random.choice(ops), random_params())
                c = {**fitness(c), "agent": agent, "round": r}
                all_cands.append(c)
                if all(iou(c["grid"], g) < 0.6 for g in archive):  # a genuinely new state
                    archive.append(c["grid"])
                if best[agent] is None or c["fit"] > best[agent]["fit"]:
                    best[agent] = c
            history.append((r, agent, best[agent], len(archive)))
    return history, all_cands
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\Users\USER\questions\Answers\Q08_MultiAgent_Drawing_Patterns> python q08_code.py
Q8  Multi-agent creative drawing (Agent A = spiral, Agent B = star)
Action space: 4 basic actions -> 7 after 3 combined actions
Search space for a motif of 6 actions: 4^6 = 4,096 -> 7^6 = 117,649 (28.7x larger)
round | agent    | best operator | fitness | novelty | aesth | archive
    1 | A-spiral | compose       |   0.703 |   0.732 | 0.525 |       2
    1 | B-star   | nest          |   0.616 |   0.499 | 0.641 |       3
    2 | A-spiral | compose       |   0.739 |   0.646 | 0.702 |       5
    2 | B-star   | compose       |   0.724 |   0.681 | 0.628 |       7
    3 | A-spiral | compose       |   0.779 |   0.684 | 0.763 |      10
    3 | B-star   | compose       |   0.735 |   0.677 | 0.661 |      13
    4 | A-spiral | compose       |   0.779 |   0.684 | 0.763 |      16
    4 | B-star   | compose       |   0.740 |   0.667 | 0.683 |      16
    5 | A-spiral | compose       |   0.779 |   0.684 | 0.763 |      18
    5 | B-star   | compose       |   0.749 |   0.669 | 0.703 |      20
    6 | A-spiral | compose       |   0.793 |   0.672 | 0.810 |      23
    6 | B-star   | compose       |   0.749 |   0.669 | 0.703 |      26
Best drawing: compose by A-spiral (round 6), fitness 0.793
Parents scored aesthetics A = 0.540, B = 0.547
charts saved to images/
PS C:\Users\USER\questions\Answers\Q08_MultiAgent_Drawing_Patterns>
```

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Figure 4. The multi-agent search function and the real terminal output.

12. Summary

> A search space is everything reachable through the available actions. Combining one action from each parent (compose, blend, nest, interleave) adds new actions. That grew the space 28.7 times and produced drawings neither parent could make. A fitness of novelty + aesthetics + balanced inheritance, plus a shared archive, guided two agents from fitness 0.62–0.70 to 0.75–0.79.

Files in this folder: `q08_code.py` (source), `q08_code.ipynb` (Colab/Jupyter version, already run), `q08_output.txt` (console output), `images/` (all figures).